S4 Pairwise Accuracy Example

Fig. S4.1 illustrates how performance on conventional metrics can mask poor *discrimination* ability, that is, the ability to distinguish vulnerable from benign code. Consider three detectors with different approaches:

1) Detector A marks vulnerable code as safe in one pair, and safe code as vulnerable in another.
2) Detector B identifies one pair correctly, but in the second pair it marks the vulnerable code as safe and incorrectly flags the safe code as vulnerable.
3) Detector C marks everything as vulnerable.

A pair is correct only when the vulnerable function is labeled vulnerable and its fixed version is labeled safe.

Detector A

	Vulnerable code	Fixed code	Pair correct?
Pair 1	FLAGGED	FLAGGED	×
Pair 2	SAFE	SAFE	×

Accuracy: 0.50
F1: 0.50
Pairwise accuracy: **0.00**

Detector B

	Vulnerable code	Fixed code	Pair correct?
Pair 1	FLAGGED	SAFE	✓
Pair 2	SAFE	FLAGGED	×

Accuracy: 0.50
F1: 0.50
Pairwise accuracy: **0.50**

Detector C: Flags everything

	Vulnerable code	Fixed code	Pair correct?
Pair 1	FLAGGED	FLAGGED	×
Pair 2	FLAGGED	FLAGGED	×

Accuracy: 0.50
F1: 0.67
Pairwise accuracy: **0.00**

Figure S4.1: A pair is correct only when vulnerable code is flagged and fixed code is not.

One can see how Detectors A and C achieve above-zero accuracy and F1 scores, with Detector C even achieving the best F1, but their pairwise accuracies are 0.